Linear Algebra - 24DS

Consolidated Assignment No. 2

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There are two types of questions; one "Mechanical" which requires you to do computations and other "Conceptual", which requires you to respond question logically using theorem and logical implications. Attempt accordingly.

I Column Space and Row Space (10 questions)

1. (**Mechanical**) For $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$, find a basis for the column space and state its dimension.

2. **(Conceptual)** Explain why the column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^m$ and the row space is a subspace of $\mathbb{R}^n$.

3. **(Mechanical)** Find bases for the column space and row space of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}.$$

4. (**Conceptual)** Is it possible that the column space of a 3×3 matrix is a line through the origin? If yes, give an example. If no, explain.

5. **(Mechanical)** For $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$, find a basis for the column space using the pivot columns from its reduced row echelon form.

6. **(Conceptual)** What is the relationship between $\text{rank}(A)$ and $\text{rank}(A^T)$? Prove it using column/row space dimensions.

7. (**Mechanical)** Let A be a 2×4 matrix. If the row space has dimension 2, what can you say about the column space? Give a concrete 2×4 matrix satisfying this and find bases for both spaces.

8. **(Conceptual)** Show that the row space and the null space are orthogonal complements in $\mathbb{R}^n$.

9. (**Mechanical)** For $A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, find bases for column space, row space, and null space. Verify the rank-nullity theorem.

10. (**Conceptual)** Prove that every vector in the column space can be written as $A\mathbf{x}$ for some $\mathbf{x}$. Conversely, if a vector can be written as $A\mathbf{x}$, it lies in the column space.

II Null Space (10 questions)

1. (**Mechanical)** Find the null space of $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. What is its dimension?

2. **(Conceptual)** Explain why the null space of a matrix is always a subspace.

3. **(Mechanical)** For $B = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$, find a basis for the null space. What is the nullity?

4. (**Conceptual)** Can the null space of a 3×3 matrix be the entire $\mathbb{R}^3$? If yes, what must the matrix be? If no, explain.

5. **(Mechanical)** Compute the null space of $C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix}$. Write the solution set in parametric vector form.

6. **(Conceptual)** If A is an $m \times n$ matrix with $\text{rank}(A) = n$, what is the null space? Why?

7. (**Mechanical)** Find all vectors in the null space of

$$D = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

(a 3×4 matrix). What is the nullity?

8. **(Conceptual)** Prove that if A is invertible, then the only solution to $A\mathbf{x} = 0$ is $\mathbf{x} = 0$. Relate to null space.

9. (**Mechanical)** For $E = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 1 \end{pmatrix}$, find the null space. Verify that null space vectors are orthogonal to each row of E.

10. (**Conceptual**) Suppose A is a 3×5 matrix with nullity 2. What is the rank? Can you determine the dimensions of the column space and row space? Explain.

III Linear Transformation and their matrices (10 questions)

1. (**Mechanical**) Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x, y) = (x + 2y, 3x + y)$. Verify that T is linear.

2. (**Conceptual**) Give an example of a function $T : \mathbb{R}^2 \to \mathbb{R}^2$ that is not linear. Explain why it fails.

3. (**Mechanical**) Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, y, z) = (x - y, y + z)$. Compute $T(1, 2, 3)$ and check additivity for two vectors of your choice.

4. (**Conceptual**) Is there a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^3$ that is onto? Why or why not?

5. (**Mechanical**) Define $T : \mathbb{R}^4 \to \mathbb{R}^4$ by $T(x_1, x_2, x_3, x_4) = (x_1, x_3, x_2, x_4)$. Show it is linear and describe geometrically what it does.

6. (**Conceptual**) If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, what must be true about $T(\mathbf{0})$? Prove it from the definition.

7. (**Mechanical**) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the projection onto the line $y = x$ along the direction of the y-axis. Find a formula for $T(x, y)$ and verify linearity.

8. (**Conceptual**) Explain why every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented by a matrix. What are the columns of that matrix?

9. (**Mechanical**) Find a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T(1, 0, 0) = (1, 1, 0)$, $T(0, 1, 0) = (1, 0, 1)$, and $T(0, 0, 1) = (0, 1, 1)$. Then compute $T(2, 3, 4)$.

10. (**Conceptual**) Can a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^4$ be one-to-one? Can it be onto? Justify using dimensions.

IV Matrices of Linear Transformations (10 questions)

1. (**Mechanical**) Find the matrix of $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (2x - y, x + 3y)$ relative to the standard basis.

2. (**Conceptual**) What is the matrix of the identity transformation $I : \mathbb{R}^n \to \mathbb{R}^n$? What about the zero transformation?

3. (**Mechanical**) Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be rotation by 90° around the z-axis (counterclockwise when looking from above). Find the matrix of T relative to the standard basis.

4. **(Conceptual)** If the matrix of $T : \mathbb{R}^2 \to \mathbb{R}^2$ is $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ relative to the standard basis, what is the matrix relative to the basis $\mathcal{B} = \{(1,1), (1,-1)\}$?

5. **(Mechanical)** For $T : \mathbb{R}^4 \to \mathbb{R}^3$ defined by $T(x_1, x_2, x_3, x_4) = (x_1+x_2, x_2+x_3, x_3+x_4)$, find its standard matrix.

6. **(Conceptual)** Explain why different bases give different matrix representations of the same linear transformation. How are these matrices related?

7. **(Mechanical)** Given the matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, describe the linear transformation $T(\mathbf{x}) = A\mathbf{x}$ geometrically. Apply it to the vector $(2, 1)$.

8. **(Conceptual)** If $T : \mathbb{R}^n \to \mathbb{R}^m$ has matrix A, what is the matrix of cT (scalar multiple) and of $T_1 + T_2$ (sum of two transformations)?

9. **(Mechanical)** Find the standard matrix of the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ that first reflects across the x-axis and then rotates by $45°$. Compute $T(1,0)$ and $T(0,1)$.

10. **(Conceptual)** Suppose $T : \mathbb{R}^3 \to \mathbb{R}^2$ and $S : \mathbb{R}^2 \to \mathbb{R}^4$ are linear with matrices A (2×3) and B (4×2). What is the matrix of $S \circ T$? What are its dimensions? Prove that matrix multiplication corresponds to composition.

V Eigen Values and Eigen Vectors (8 questions)

1. **(Mechanical)** For $E = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$, find eigenvalues and eigenvectors. Verify that eigenvectors corresponding to distinct eigenvalues are orthogonal. Note orthogonal vectors are perpendicular, which make 90 degree angle and their dot product is zero i.e. $\mathbf{u}.\mathbf{v} = 0$.

2. **(Conceptual)** Let A be a 2×2 matrix with eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$. What are the eigenvalues of A^2? Of A^{-1} (if invertible)? Of $A + 3I$?

3. **(Conceptual)** Show that a 2×2 real matrix with a repeated eigenvalue λ is diagonalizable if and only if it is already λI.

4. **(Conceptual)** Suppose v is an eigenvector of both A and B. Is v necessarily an eigenvector of AB? Prove or give a counterexample.

5. **(Mechanical)** Compute eigenvalues and eigenvectors for

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

6. **(Conceptual)** If A is a 3×3 matrix with eigenvalues $1, 2, 3$, what are the eigenvalues of A^T? What about $A^2 - 2A + I$?

7. **(Mechanical)** Find the eigenvalues and a basis for each eigenspace of

$$D = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

   Is D defective? Note that matrix A is called defective if their all eigenvectors are not linearly independent, means non-diagonalizable.

8. **(Conceptual)** Prove that if λ is an eigenvalue of an orthogonal matrix Q, then $|\lambda| = 1$. Note that matrix A is called orthogonal if $A^T = A^{-1}$

VI Diagonalization (6 questions)

1. **(Mechanical)** Diagonalize the matrix $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. That is, find P and D such that $A = PDP^{-1}$.

2. **(Mechanical)** Find P and D for $C = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$ and compute C^{10} using the diagonalization.

3. **(Conceptual)** Can a non-invertible 2×2 matrix be diagonalizable? If yes, give an example. If no, explain why.

4. **(Mechanical)** Determine whether $B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ is diagonalizable. If so, diagonalize it.

5. **(Conceptual)** Suppose A and B are diagonalizable 2×2 matrices with the same eigenvectors. Does it follow that $AB = BA$? Explain.

6. **(Mechanical)** Diagonalize the symmetric matrix

$$D = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

   with an orthogonal matrix P (i.e., find P orthogonal and D diagonal).